

```

<!DOCTYPE html>
<html>
<head>

  <title>D3 + Labels + Cluster LayerSupport</title>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.3/dist/leaflet.css" />
  <!--
    #SILAS you are loading leaflet twice below - REMOVE 1
  -->
  <script src="https://unpkg.com/leaflet@1.9.3/dist/leaflet.js"></script>
  <!-- <script src="https://unpkg.com/leaflet@1.9.3/dist/leaflet.js"></script> -->
  <link rel="stylesheet" href="./leaflet-mapwithlabels.css" />
  <script src="./leaflet-mapwithlabels.js"></script>
  <link rel="stylesheet" href="./leaflet.awesome-markers.css">
  <link href="http://netdna.bootstrapcdn.com/font-awesome/4.0.0/css/font-awesome.css" rel="stylesheet">
  <link href="http://netdna.bootstrapcdn.com/bootstrap/3.0.0/css/bootstrap.min.css" rel="stylesheet">
  <link rel="stylesheet" href="https://unpkg.com/leaflet-control-geocoder/dist/Control.Geocoder.css" />
<script src="https://unpkg.com/leaflet-control-geocoder/dist/Control.Geocoder.js"></script>
<link
  rel="stylesheet"
  href="https://unpkg.com/leaflet-geosearch@3.0.0/dist/geosearch.css"
/>
<script src="https://unpkg.com/leaflet-geosearch@latest/dist/bundle.min.js"></script>
<style>

  body { margin: 0; padding: 0; }
  #map_div { height: 600px; width: 100%; }
  .info { padding: 6px 8px; font: 14px/16px Arial; background: white; border-radius: 5px; box-shadow: 0 0
15px rgba(0,0,0,0.2); }
  .legend i { width: 18px; height: 18px; float: left; margin-right: 8px; opacity: 0.7; }
  html { height: 100% }
  body { height: 100%; margin: 0; padding: 0 }
  #map-canvas { height: 100% }
  html, body, #map-canvas { height: 100%; margin: 0; padding: 0; background: #f0f0f0; }

  path.neighborhood {
    cursor: pointer;
    transition: fill-opacity 0.3s ease;
  }

.leaflet-marker-icon, .leaflet-marker-shadow, .leaflet-image-layer, .leaflet-pane > svg path, .leaflet-tile-container
{
  pointer-events: visible;
}

/* High-visibility label styling - EXTRA BOLD */
.d3-label {
  font-family: "Helvetica Neue", Helvetica, Arial, sans-serif;
  font-weight: 900;
  fill: #111;
  text-anchor: middle;
  pointer-events: none;
  text-shadow: 2px 2px 0px #fff, -2px -2px 0px #fff, 2px -2px 0px #fff, -2px 2px 0px #fff;
}
html, body, #map_div {

```

```
height: 100% !important;
width: 100% !important;
margin: 0;
padding: 0;
-webkit-backface-visibility: hidden; /* Fixes "shimmer" on some browsers */
}
```

/* Add this to your map page styles */

```
.leaflet-zoom-animated {
  display: block !important; /* Forces the SVG layer to stay rendered */
}
```

```
.leaflet-tile-container {
  transition: opacity 0.2s linear; /* Makes tile loading less jarring */
}
```

/* Ensure the labels and polygons stay on the same rendering plane */

```
.leaflet-pane {
  z-index: 400;
}
```

/* Prevents the SVG renderer from clipping polygons during a resize */

/* Force the SVG container to always fill the available space */

/* Force the SVG container to fill the 100% space of the expanded iframe */

```
.s{
  width: 100% !important;
  height: 100% !important;
  transform-origin: 0 0 !important;
  overflow: visible !important;
}
```

/* Ensure labels don't "stutter" during the move */

```
.leaflet-label {
  will-change: transform, top, left;
  transition: none !important;
}
```

/* Ensure the path (polygon) doesn't disappear or flicker */

/* Keep the polygon fill stable */

```
path.leaflet-interactive {
  /* Prevents borders from getting thick and overlapping during zoom */
  vector-effect: non-scaling-stroke !important;
  stroke-width: 1px !important;
}
```

</style>

<link rel="stylesheet" href="./MarkerCluster.css" />

<link rel="stylesheet" href="./MarkerCluster.Default.css" />

<script src="./leaflet.markercluster-src.js"></script>

<!-- #SILAS <script src="./leaflet.markercluster.freezeable.js"></script> -->

<!-- After Leaflet script -->

<script src="https://unpkg.com/leaflet.markercluster.freezeable@1.0.0/dist/leaflet.markercluster.freezeable.js"></script>

<!-- #SILAS This looks like the wrong file name to me -->

<!-- <script src="./LDeflate (1).js"></script> -->

<!-- #SILAS use the online version -- this is from the github site -->

```

<script src="https://unpkg.com/Leaflet.Deflate/dist/L.Deflate.js"></script>
<script src="/.leaflet.awesome-markers.js"></script>
<!-- #SILAS I couldn't find the local version so I'm using the online version below -->
<!-- <script src="/.leaflet.markercluster.layersupport.js"></script> -->
<!-- After Leaflet and Leaflet.markercluster scripts -->
<script src="https://unpkg.com/leaflet.markercluster.layersupport@2.0.1
/dist/leaflet.markercluster.layersupport.js"></script>

</head>
<body>
  <!--
  #SILAS:
  -- you have two divs with id map_canvas -- REMOVE ONE OF THEM
  -- you are adding the /.leaflet.markercluster-src.js THREE times -- REMOVE THE SECOND AND THIRD
TIMES

  -->
  <div id="map_canvas"></div>
  <!-- #SILAS this next line is not a script it is just a web site -->
  <!-- <script src="https://d3js.org"></script> -->

  <!-- <script src="/.leaflet.markercluster-src.js"></script> -->
  <script src="L.D3SvgOverlay.min.js"></script>
  <!-- #SILAS - you're already loading this with a different name leaflet-mapwithlabels.js (there's a HYPHEN
instead of a period)-->
  <!-- <script src="/.leaflet.mapwithlabels.js"></script> -->
  <!-- div id="map_canvas"></div -->

  <!-- #SILAS you already load leaflet.js in the head - REMOVE THIS NEXT ONE -->
  <!-- <script src="https://cdnjs.cloudflare.com/ajax/libs/leaflet/0.7.7/leaflet.js"></script> -->
  <script src="https://cdnjs.cloudflare.com/ajax/libs/d3/3.4.9/d3.min.js"></script>

  <!-- <script src="/.leaflet.markercluster-src.js"></script> -->
  <!-- Second: The LayerSupport sub-plugin (depends on the one above) -->

  <!-- Third: D3 Overlay -->
  <script src="L.D3SvgOverlay.min.js"></script>
  <!-- Fourth: Your local Label plugin (Fix path if needed) -->

  <!-- #SILAS - you are loading this above already -- REMOVE THIS ONE-->
  <!-- <script src="/.leaflet.mapwithlabels.js"></script> -->

  <!-- #SILAS - import the cambridgeSomervilleNeighborhoods with a var so you can use this with the marker
cluster-->
  <!-- this stores the cambridgeSomervilleNeighborhoods to a var named
"cambridgeSomervilleNeighborhoods"-->
  <script src="/cambridgeSomervilleNeighborhoods.geojson.js"></script>

<script>
  // #SILAS -- adding the polycolor function here. This creates the random colors that are given to each
neighborhood
  // -- if you search polycolor in this html, you'll see where this function is called
  function polyColor(f) {
    let p = f.properties.Nepesseg;
    let r = Math.random() * 150 + 100;

```

```

    let g = Math.random() * 150 + 100;
    let b = Math.random() * 150 + 100;
    return { fillColor: `rgb(${r},${g},${b})`, fillOpacity: .7, color: 'black', opacity: 1, weight: 1 };
}

```

// 1. Initialize Map

```

var map;
try {
  map = L.mapWithLabels('map-canvas', {
    center: [42.3840352, -71.1134377],
    zoom:13
  });
} catch (e) {
  console.warn("mapWithLabels plugin not found, using standard Leaflet map.");
  map = L.map('map-canvas').setView([42.3840352, -71.1134377], 13);
}

```

```

L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
  attribution: '&copy; OpenStreetMap contributors'
}).addTo(map);

```

```

var neighborhoodData = [];

```

// 2. Define the D3 Overlay (Shapes + Labels + Interactions)

```

var d3Overlay = L.d3SvgOverlay(function (sel, proj) {

  // --- DRAW NEIGHBORHOOD SHAPES ---
  var paths = sel.selectAll('path').data(neighborhoodData);

  paths.enter()
    .append('path')
    .attr('class', 'neighborhood')
    .attr('stroke', 'black')
    .attr('fill', function() { return d3.hsl(Math.random() * 360, 0.7, 0.5); })
    .attr('fill-opacity', '0.5')
  // HOVER EFFECTS
  .on('mouseover', function(d) {
    d3.select(this)
      .transition().duration(150)
      .attr('fill-opacity', '0.8')
      .attr('stroke-width', 3 / proj.scale);
  })
  .on('mouseout', function(d) {
    d3.select(this)
      .transition().duration(150)
      .attr('fill-opacity', '0.5')
      .attr('stroke-width', 1 / proj.scale);
  })
  // CLICK TO ZOOM
  .on('click', function(d) {
    map.fitBounds(L.geoJson(d).getBounds());
  });
});

```

// Sync path projection to map zoom/pan

```

paths.attr('d', proj.pathFromGeojson)

```

```

.attr('stroke-width', 1 / proj.scale);

// --- DRAW LABELS ---
var labels = sel.selectAll('text').data(neighborhoodData);

labels.each(function(d) {
  // Find center of each polygon for label placement
  var center = L.geoJson(d).getBounds().getCenter();
  var point = proj.latLngToLayerPoint(center);
  d3.select(this)
    .attr('x', point.x)
    .attr('y', point.y)
    .style('font-size', (14 / proj.scale) + 'px');
});
});

// 3. LOAD DATA AND ACTIVATE
// IMPORTANT: Ensure this filename matches your .geojson file exactly!
d3.json("./somervilleNeighborhoods (2).geo.json", function(error, data) {
  if (error) {
    console.error("GeoJSON failed to load. Check file path or use a local server.", error);
    return;
  }

  neighborhoodData = data.features;

  // Add the D3 layer to the map now that we have data
  d3Overlay.addTo(map);

  console.log("Map initialized with", neighborhoodData.length, "neighborhoods.");

});

var mcgLayerSupportGroup = L.markerClusterGroup.layerSupport();

// Create separate layer groups for different types of markers
var stores = L.layerGroup();
var homes = L.layerGroup();
var bookstores = L.layerGroup();
var WestCambridge = L.layerGroup();

// Check these layer groups into the MCG Layer Support

mcgLayerSupportGroup.checkIn([stores, homes, bookstores, WestCambridge]);

// Add markers to their respective layer groups
L.marker([42.38985, -71.1185]).bindPopup('StarMarket').addTo(stores);
L.marker([42.39392, -71.125408]).bindPopup('Pemberton').addTo(stores);
L.marker([42.367843, -71.099929]).bindPopup('229 Harvard').addTo(homes);
L.marker([42.401424, -71.13354]).bindPopup('26 Murray Hill Road').addTo(homes);
L.marker([42.3817724, -71.0872455]).bindPopup('All She Wrote Books').addTo(bookstores);
L.marker([42.3812783, -71.1008173]).bindPopup('Side Quest Books & Games').addTo(bookstores);
L.marker([42.372541, -71.116363]).bindPopup('Harvard Book Store').addTo(bookstores);

```

```
L.marker([42.3873306,-71.1214895]).bindPopup('Porter Square Books').addTo(bookstores);
L.marker([42.3812973,-73.5779292]).bindPopup('Bryn Mawr Bookstore').addTo(bookstores);
L.marker([42.3738631,-71.1242451]).bindPopup('Lovestruck Books').addTo(WestCambridge);
```

```
mcgLayerSupportGroup.addTo(map);
```

```
// Add the layer groups to the map to display them (they will be clustered automatically)
```

```
stores.addTo(map);
homes.addTo(map);
WestCambridge.addTo(map);
bookstores.addTo(map);
```

```
// Add the standard Leaflet layers control
```

```
var overlays = {
  "Grocery Stores": stores,
  "Homes": homes,
  "Bookstores": bookstores,
  "West Cambridge": WestCambridge
};
L.control.layers(null, overlays).addTo(map);
```

```
var markers = L.markerClusterGroup({spiderfyOnMaxZoom: false, showCoverageOnHover: false,
zoomToBoundsOnClick: false});
markers.on('clusterclick', function (a) {
  a.layer.zoomToBounds({padding: [60, 60]});
});
```

```
// Store the layer so we can find Davis Square later
```

```
// #SILAS - CHANGE THIS TO the cambridgeSomervilleNeighborhoods
```

```
// -- replacing somervilleNeighborhoods
```

```
const neighborhoodLayer = L.geoJSON(cambridgeSomervilleNeighborhoods, {
  label: l => l.feature.properties.NAME,
  labelPos: 'cc',
  labelStyle: { color: 'red', fontWeight: 'bold', whiteSpace: 'normal', minWidth: '120px', textAlign: 'center' },
  labelPriority: 5e4,
  style: polyColor
}).addTo(map).bindPopup(l => `<b>${l.feature.properties.NAME}</b>`);
```

```
// Source - https://stackoverflow.com/a/25806894
```

```
// Posted by StudioTime, modified by community. See post 'Timeline' for change history
```

```
// Retrieved 2026-03-02, License - CC BY-SA 3.0
```

```
// 2. Global variable for your polygon layer
```

```
var region; // Global variable for your combined GeoJSON layer
```

```
// 1. Listen for the message from the parent
```

```
window.addEventListener('message', (event) => {
  if (event.data && event.data.action === 'sync-resize-fly') {
    const vc = event.data.vc;
    if (vc) {
      console.log("Flying to location:", vc);
      runSearch(vc);
    }
  }
});
```

```
}  
});
```

```
// 2. Load the single file containing BOTH cities  
fetch('/somervilleNeighborhoods (2).geo.json')  
  .then(response => response.json())  
  .then(data => {  
    region = L.geoJSON(data, {  
      style: { color: 'blue', weight: 2, fillOpacity: 0.1 }  
    }).addTo(map);  
  })  
  .catch(err => console.error("Error loading GeoJSON:", err));
```

```
// 3. Geosearch function  
function runSearch(vc) {  
  if (!region) return console.error("Region data not loaded.");  
  
  const provider = new window.GeoSearch.OpenStreetMapProvider();  
  
  const searchQuery = `${vc}, Somerville or Cambridge, MA`;  
  
  provider.search({ query: searchQuery }).then(results => {  
    if (results.length === 0) {  
      console.error("No results found for:", searchQuery);  
      return;  
    }  
  
    // OSM returns an array; we want the most relevant first one  
    const result = results[0];  
    const latlng = L.latLng(result.y, result.x);  
  
    console.log("Found Coordinates:", latlng.toString(), "for", result.label);  
  
    let isInside = false;  
    region.eachLayer(function(layer) {  
      if (isInside) return;  
      if (isMarkerInsidePolygon(latlng, layer)) {  
        isInside = true;  
      }  
    });  
  
    if (isInside) {  
      map.flyTo(latlng, 17);  
      L.marker(latlng).addTo(map)  
        .bindPopup(`<b>${vc}</b><br>${result.label}`)  
        .openPopup();  
    } else {  
      console.warn("Location is outside the GeoJSON boundary:", result.label);  
      // DEBUG: Fly there anyway to see where the map thinks the museum is  
      map.flyTo(latlng, 15);  
    }  
  });  
}
```

</script>

</body>

</html>